Fixed-Income Securities: Concise Summary

Navid

March 3, 2026

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List of Symbols

Notation	Description
$F_t(T_1, T_2)$	Forward rate at time t for operations between T_1 and T_2 (first argument is dropped when $t = 0$)
$f(t, T)$	Instantaneous forward rate at time t for operations at time T
$\Phi(\cdot)$	Standard normal cumulative distribution function (cdf)
$i_t(T)$	Money market (simple) rate at time t with horizon T
K_f	Forward swap rate
$\chi^2(x, y, z)$	Cumulative distribution function of the non-central chi-square law with y degrees of freedom and centrality parameter z evaluated at x
$P(t, T)$	Discount factor, zero-coupon bond price
r_t	Instantaneous risk-free rate (stochastic process)
τ	Interest rate option or swap periodicity, or Nelson-Siegel-Svensson curvature parameter
θ	Nelson-Siegel-Svensson curvature parameter
W_t	Brownian motion under the physical measure
$zcy(t, T)$	Zero-coupon yield
Z_t	Brownian motion under the Equivalent Martingale Measure

1 Fundamentals

Compounding

Discrete: $\prod_{j=0}^{n-1}(1 + r_j)$ $[r_j =$ spot rate at beginning of period $j]$

Continuous: $\exp\left(\int_0^T r_u \, du\right)$ $[\{r_u\}_{u \geq 0} =$ stochastic process for instantaneous risk-free rate]

Discounting

Discrete: $\prod_{j=0}^{n-1}(1 + r_j)^{-1}$ [discount one unit from time T to current time]

Continuous: $\exp\left(-\int_0^T r_u \, du\right)$

Discount Factor (Zero-Coupon Bond Price)

$$P(t, T) = \mathbb{E}_t^Q \left[\exp\left(-\int_t^T r_u \, du\right)\right]$$

Properties: $0 < P(t, T) < 1$, decreasing and convex in T

Zero-Coupon Yield (spot rate)

$$zcy(t, T) = -\frac{1}{T - t} \ln P(t, T)$$

Forward Rates

for Simple rates: $F(T_1, T_2) = \frac{1}{T_2 - T_1}\left(\frac{1+i(T_2)T_2}{1+i(T_1)T_1} - 1\right)$
for continous rates: $F(T_1, T_2) = \frac{1}{T_2 - T_1} \ln \frac{P(0,T_1)}{P(0,T_2)}$
As function of zero-coupon yields: $F(T_1, T_2) = \frac{zcy(0,T_2)T_2 - zcy(0,T_1)T_1}{T_2 - T_1}$

Instantaneous Forward Rate

Definition: $f(T) = \lim_{T_2 \to T} F(T, T_2) = -\frac{\partial \ln P(0,T)}{\partial T}$
Relation to discount factor: $P(0, T) = \exp\left(-\int_0^T f(u)\, du\right)$

Money Market Instruments (use simple, proportional rate.)

Short maturity (below 1 year), no interim payments. eg: Treasury bills, commercial papers, banker's acceptances, municipal notes.

Discount basis: $P = M(1 - i(T)T)$, yield: $y_d = \frac{M-P}{MT}$ [used in North America, UK, EU]
Money-market basis: $P = \frac{M}{1+i(T)T}$, yield: $y_m = \frac{M-P}{PT}$ [used in Japan]
Difference: money-market basis accounts for time value of interests, more favorable to lender than discount basis.

Repurchase agreements (repos): sell securities with commitment to repurchase. Financing rate implied by sell/repurchase price difference. Close to central bank overnight rate, affected by liquidity and collateral demand.

Bonds

Zero-coupon bond (STRIPS): Value $P(0, T)$ for unit face value at maturity T.
Coupon-bearing bond (par issuance, semi-annual payments):

$$\sum_{j=1}^{2T} \frac{c}{2} M P(0, \frac{j}{2}) + M P(0, T)$$

[c = coupon rate, M = face value, T = maturity]
Re-evaluation between coupon dates accounts for accrued interest.
Embedded options: callable (can be redeemed by the issuer at pre-specified price/period), convertible (can be converted by the holder into a pre-specified fraction of equity, during a given period.), callable convertible (combines call and conversion features), putable (can be redeemed by the holder at pre-specified price/period).

Forward Rate Agreement (FRA)

OTC, notional M, effective date T_1, termination date T_2, reference rate i_t, fixed rate R.
Buyer cash flow at T_1: $\frac{M(T_2 - T_1)(i_{T_1}(T_2) - R)}{1 + i_{T_1}(T_2)(T_2 - T_1)}$ — seller cash flow is negative of buyer's.
If initial value nil: $R = F(T_1, T_2)$
No principal exchange. Flexible hedging instrument. Buyer can borrow at rate R for period $[T_1, T_2]$.

FRA Futures

Montreal Exchange - CORRA futures (1-month, 3-month maturity): quoted as $100 - R$ where R = compounded daily CORRA. Contract size: \$2500 $\times$ $(100 - R)$ per contract [\$25 per basis point — 1 bp =0.0001].
CME group - SOFR futures (1-month, 3-month maturity): similar to CORRA. Available up to 10 years ahead.

Hedging by selling h futures (3-month, M million loan at variable rate): $h = \frac{M \text{ mil.} \times 1 \text{ bp} \times 91/365}{25} \approx M$

Treasury Bill Futures

CME group: quoted as $100 - R$ where R = yield on 13-week T-bill. Contract size: $\$2500 \times (100 - R)$ [\$25 per basis point]. Limited to one year of traded reference quarters.

Treasury Bond Futures

Underlying and Delivery: Montreal Exchange - 4 maturities (2.5, 10, 20, 30 years). Government bond with \$100,000 nominal and 6% coupon. Conversion factor (CF) adjusts delivered bond yields to 6%. Physical delivery.
Delivery Options: - Timing option: seller delivers any day in delivery month - End-of-month option: few business days between last trading day and delivery date - Quality option: seller chooses bond from pool of deliverable bonds
Invoice Price:

$$IP_t = H_{t,T} \times CF$$

where $H_{t,T}$ = futures price at sale, CF = conversion factor for delivered bond
Forward Price (Cash & Carry):

$$FP_{t,T} = (SP_t + AC_t)\left(1 + r\frac{T-t}{365}\right) - AC_T - \frac{c}{2}\left(1 + r\frac{T-s}{365}\right)$$

where SP_t = spot price at time t, AC_t = accrued interest at time t, AC_T = accrued interest at delivery T, r = implicit financing rate, c = coupon, s = last coupon date before T
Basis After Carry:

$$BAC_t = FP_{t,T} - IP_t$$

Cheapest-to-deliver (CTD) bond: minimize BAC_t at time t (Maximize r^* to find CTD. Higher r^* favors seller over holding bond.)
Implied Repo Rate:

$$r^* = \frac{H_{t,T} \times CF - SP_t + AC_T - AC_t - c/2}{(SP_t + AC_t)\frac{T-t}{365} - \frac{c}{2}\frac{T-s}{365}}$$

Options on Bond Futures

American options written on bond futures (Montreal: 2-year, 5-year, 10-year). Option maturities = 3 months preceding futures maturity. Weekly options also available.

Interest Rate Options (Caps & Floors)

OTC derivatives. *Cap* = chain of European call options ("caplets"). *Floor* = chain of European put options ("floorlets").
Contract: notional M, reference rate $i_t(\cdot)$, strike K, maturity T, settlement dates $t_1, t_2, \ldots, t_n = T$.
Period: $\tau = t_j - t_{j-1}$ (typically quarterly)
Caplet payoff at t_j (j=1,...,n):

$$M\tau(i_{t_{j-1}}(t_j) - K)^+$$

This is payoff of put on $P(t_{j-1}, t_j)$ with: strike $K' = \frac{1}{1+K\tau}$, nominal $M' = M(1 + K\tau)$, maturity t_{j-1}

$$\text{Cap}(M, K) = \sum_{j=\alpha+1}^{n} M'\text{Put}(P(t_{j-1}, t_j), K')$$

$$\text{Floor}(M, K) = \sum_{j=\alpha+1}^{n} M'\text{Call}(P(t_{j-1}, t_j), K')$$

where t_α = first reference rate reset date, $t_{\alpha+1}$ = first settlement date.

Interest Rate Swaps

Collar: Buy cap at K_2, sell floor at K_1 (where $K_1 < K_2$). Net position each quarter t_j:

$$-M\tau i_{t_{j-1}}(t_j) + M\tau(i_{t_{j-1}}(t_j) - K_2)^+ - M\tau(K_1 - i_{t_{j-1}}(t_j))^+$$

When $K_1 = K_2 = K$, collar becomes *interest rate swap.* No-arbitrage:

$$\text{Cap}(M, i, K) - \text{Floor}(M, i, K) = \text{Payer swap}(M, i, K)$$

$$\text{Floor}(M, i, K) - \text{Cap}(M, i, K) = \text{Receiver swap}(M, i, K)$$

Payer swap (fixed rate payer): pays K per period, receives floating $i_{t_{j-1}}(t_j)$

$$\text{Payer swap}(M, i, K) = M\tau \sum_{j=\alpha+1}^{n} P(0, t_j)(F(t_{j-1}, t_j) - K)$$

where $F(t_{j-1}, t_j) = \frac{1}{\tau}\left(\frac{P(0,t_{j-1})}{P(0,t_j)} - 1\right)$ = simply compounded forward rate
Simplified form:

$$\text{Payer swap}(M, i, K) = M(P(0, t_\alpha) - P(0, t_n)) - M\tau K \sum_{j=\alpha+1}^{n} P(0, t_j)$$

Receiver swap (fixed rate receiver): receives K per period, pays floating

$$\text{Receiver swap}(M, i, K) = M(P(0, t_n) - P(0, t_\alpha)) + M\tau K \sum_{j=\alpha+1}^{n} P(0, t_j)$$

Forward swap rate: Fixed rate for which payer and receiver swaps have zero value

$$K_f = \frac{P(0, t_\alpha) - P(0, t_n)}{\tau \sum_{j=\alpha+1}^{n} P(0, t_j)}$$

Swaptions

European option on interest rate swap. *Payer swaption*: right to enter payer swap. *Receiver swaption*: right to enter receiver swap. Swaption maturity = t_α (swap's first reset date). Tenor = $t_n - t_\alpha$.

Payer swaption payoff:

$$M\tau \sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)^+$$

Thus, a swaption can be viewed as an option on a coupon-bearing bond.
Compared to caps/floors: swaption value < cap/floor value since:

$$\left(\sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)\right)^+ \leq \sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)^+$$

The positive part cannot be "pushed inside" the sum.

note: January (31), February (28), March (31), April (30), May (31), June (30), July (31), August (31), September (30), October (31), November (30), December (31)

2 Yield Curve Smoothing

2.1 Money Market to Zero-Coupon Bonds

Money market instruments quoted on discount basis. For t-maturity money market rate $R_M(t)$ (day-count: Actual/365 Canada, Actual/360 US):

$$P(0, t) = 1 - R_M(t) \cdot t$$

2.2 Bootstrapping Method

Long-term swap/bond rates $R_S(T)$ (bullet redemption, semi-annual coupons):

$$100 = \sum_{i=1}^{T} \frac{100 \times R_S(T)}{2} P\left(0, \frac{i}{2}\right) + 100 \times P(0, T)$$

Solve for discount factors $P(0, i)$ recursively:
Initial: $P(0, 1/2)$ and $P(0, 1)$ from money market.
Next: From 2-year OIS rate/Bond yield $R_S(2)$, Solve for $P(0, 2)$. for $P(0, 1.5)$, linearly interpolate between $P(0, 1)$ and $P(0, 2)$. then turn the Ps into ZCYs.

2.3 Nelson-Siegel-Svensson Smoothing

$$\text{zcy}_{\text{NSS}}(T) = a + b\left(\frac{1 - e^{-T/\tau}}{T/\tau}\right) + c\left(\frac{1 - e^{-T/\tau}}{T/\tau} - e^{-T/\tau}\right) + d\left(\frac{1 - e^{-T/\theta}}{T/\theta} - e^{-T/\theta}\right)$$

Parameters: a = long-end level; b = short-end/slope; c, d = two curvatures; τ, θ = maturity of two curvatures.

$$\text{Instantaneous forward rate:} f_{\text{NSS}}(T) = a + \frac{e^{-T/\tau}}{T}(b\tau + cT) + \frac{e^{-T/\theta}}{T} dT$$

Fitting: set $w_i = 1$,for no weighting. $w_i = \frac{1/t_i}{\sum_{k=1}^{n} 1/k}$ (short-end emphasis) or $w_i = \frac{i}{\sum_{k=1}^{n} k}$ (long-end emphasis).

$$\min_{a,b,c,d,\tau,\theta} \sum_{i=1}^{n} (\text{zcy}_{\text{NSS}}(t_i) - \text{zcy}(t_i))^2 w_i^2$$

3 One-factor Models

3.1 Vasicek Model (equilibrium model)

EMM dynamics of short rate:

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t$$

κ = speed of mean reversion; θ = long-run mean; σ = volatility. all must be positive. Negative rates possible but unlikely with reasonable parameters.

$$r_T = r_t e^{-\kappa(T-t)} + \theta(1 - e^{-\kappa(T-t)}) + \sigma \int_t^T e^{-\kappa(T-u)} dW_u$$

$$\mathbb{E}_t[r_T] = \theta + (r_t - \theta)e^{-\kappa(T-t)}$$

$$\mathrm{Var}_t[r_T] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa(T-t)})$$

Physical measure:

$$dr_t = \kappa^P(\theta^P - r_t)dt + \sigma dZ_t$$

where $\kappa^P = \kappa - \sigma\lambda_1$ and $\theta^P = \frac{\kappa\theta + \sigma\lambda_0}{\kappa - \sigma\lambda_1}$.
Stationarity requires: $\lambda_1 < \kappa/\sigma$.

$$P_{vk}(t,T) = a_{vk}(t,T)\exp[-b_{vk}(t,T)r_t]$$

$$\mathrm{zcy}_k(t,T) = -\frac{1}{T-t}\ln[a_{vk}(t,T)) - b_{vk}(t,T)r_t]$$

where

$$a_{vk}(t,T) = \exp\left[\left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(b_{vk}(t,T) - T + t) - \frac{\sigma^2}{4\kappa}b_{vk}(t,T)^2\right]$$

$$b_{vk}(t,T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa}$$

Option on zero-coupon bond: European call on nominal \$1 zero-coupon bond (maturity T) with strike K and option maturity T_{call}:

$$\mathrm{call}_{vk}(P,K) = P_{vk}(0,T)\Phi(h) - KP_{vk}(0,T_{\text{call}})\Phi(h - \sigma_p)$$

$$h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{vk}(0,T)}{KP_{vk}(0,T_{\text{call}})}$$

$$\sigma_p = \frac{\sigma}{\kappa}(1 - e^{-\kappa(T-T_{\text{call}})})\sqrt{\frac{1 - e^{-2\kappa(T_{\text{call}}-t)}}{2\kappa}}$$

$$\mathrm{put}_{vk}(P,K) = KP_{vk}(0,T_{\text{put}})\Phi(\sigma_p - h) - P_{vk}(0,T)\Phi(-h)$$

Interpretation (Black-Scholes analogy): Bond value is log-normally distributed. σ_p = volatility of zero-coupon bond returns; $\Phi(h)$ = call delta; $\Phi(h - \sigma_p)$ = probability (under EMM) call is exercised.

Option on coupon bond: Coupon bond replicates portfolio of zero-coupon bonds: $P_c(0, \{T_i\}) = \sum_{i=1}^n c_i P(0, T_i)$.
European call (strike K, maturity T_{call}) terminal payoff: $(\sum_{i=1}^n c_i P(T_{\text{call}}, T_i) - K)^+$.
Jamshidian decomposition: Find critical short rate r^* where coupon bond equals strike: $\sum_{i=1}^n c_i P(T_{\text{call}}, T_i, r^*) = K$. Then:

$$\text{call}(P_c(0, \{T_i\}), K) = \sum_{i=1}^n c_i \,\text{call}(P(0, T_i), K_i)$$

where $K_i = P(T_{\text{call}}, T_i, r^*)$.
Swaptions: Payer swaption: right to enter payer swap (fixed rate K, first reset t_α, settlements $t_{\alpha+1}, ..., t_n$).
Rewrite as pseudo-coupon bond: $c_i = K\tau$ for $i = \alpha+1, ..., n-1$; $c_n = 1 + K\tau$.
Find r^* (at-the-money): $\sum_{i=\alpha+1}^n c_i P(t_\alpha, t_i, r^*) = 1$.

$$\text{payer swaption}(M, K, t_\alpha) = M \sum_{i=\alpha+1}^n c_i \,\text{put}(P(t_\alpha, t_i), K_i)$$

$$\text{receiver swaption}(M, K, t_\alpha) = M \sum_{i=\alpha+1}^n c_i \,\text{call}(P(t_\alpha, t_i), K_i)$$

where $K_i = P(t_\alpha, t_i, r^*)$. Use zero-coupon bond options to evaluate each term.

3.2 Cox-Ingersoll-Ross (CIR) Model (equilibrium model)

EMM dynamics:

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$$

Parameters κ, θ, σ as in Vasicek. Key feature: **square-root process** ensures $r_t \geq 0$ if $r_0 \geq 0$ and Feller condition holds: $2\kappa\theta > \sigma^2$.
Distribution: r_t follows non-central chi-square (scaled). For simulation: $r_t = Y/b_t$ where $b_t = \frac{4\kappa}{\sigma^2(1-e^{-\kappa t})}$ and $Y \sim \chi^2_\nu(\lambda_c)$ non-central chi-square with $\nu = 4\kappa\theta/\sigma^2$ degrees of freedom and non-centrality $b_t r_0 e^{-\kappa t}$.
Physical measure:

$$dr_t = \kappa^P(\theta^P - r_t)dt + \sigma\sqrt{r_t}dZ_t$$

where $\kappa^P = \kappa - \sigma\lambda$ and $\theta^P = \frac{\kappa\theta}{\kappa - \sigma\lambda}$. Stationarity: $\lambda < \kappa/\sigma$.
Zero-coupon bond value (closed form):

$$P_{\text{cir}}(t, T) = a_{\text{cir}}(t, T) \exp[-b_{\text{cir}}(t, T) r_t]$$

$$a_{\text{cir}}(t, T) = \left(\frac{2\gamma e^{\frac{\kappa+\gamma}{2}(T-t)}}{(\kappa+\gamma)(e^{\gamma(T-t)} - 1) + 2\gamma} \right)^{\frac{2\kappa\theta}{\sigma^2}}$$

$$b_{\text{cir}}(t, T) = \frac{2(e^{\gamma(T-t)} - 1)}{(\kappa+\gamma)(e^{\gamma(T-t)} - 1) + 2\gamma}$$

$$\gamma = \sqrt{\kappa^2 + 2\sigma^2}$$

$$\text{zcy}(t,T) = -\frac{1}{T-t}[\ln(a_{\text{cir}}(t,T)) - b_{\text{cir}}(t,T)r_t]$$

Option on zero-coupon bond (CIR): European call with strike K and maturity T_{call}:

$$\text{call}_{\text{cir}}(P,K) = P_{\text{cir}}(0,T)\chi^2_\xi(\cdot) - KP_{\text{cir}}(0,T_{\text{call}})\chi^2_\psi(\cdot)$$

where χ^2 denotes non-central chi-square distribution.

3.3 Hull-White Model (arbitrage model)

$$dr_t = \kappa(\varphi(t) - r_t)dt + \sigma dW_t$$

where $\varphi(t)$ (deterministic function of time) calibrates perfectly to the observed yield curve at $t = 0$. At $t = 0$, model fits the yield curve exactly: $P_{\text{hw}}(0,T) = P^m(0,T)$ (observed prices).
Zero-coupon bond value: Same affine structure as Vasicek:

$$P_{\text{hw}}(t,T) = a_{\text{hw}}(t,T)\exp[-b_{\text{hw}}(t,T)r_t]$$

where $b_{\text{hw}}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$ (identical to Vasicek) and

$$a_{\text{hw}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp[b_{\text{hw}}(t,T)f^m(0,t) - \frac{\sigma^2}{4\kappa}b_{\text{hw}}(t,T)^2(1-e^{-2\kappa t})]$$

All other formulas follow Vasicek structure.

3.4 Brigo-Mercurio Model (arbitrage model)

Shifted CIR model: Adds deterministic shift to CIR to fit yield curve to fit at $t = 0$:

$$dx_t = \kappa(\theta - x_t)dt + \sigma\sqrt{x_t}dW_t$$

$$r_t = x_t + \varphi(t)$$

Model fits yield curve when: $\varphi(t) = f^m(0,t) - f^{cir}(0,t)$
Warning: Unlike CIR, Brigo-Mercurio does **not guarantee positive rates**. positivity is gauranteed if $\varphi(t) > 0$ for all t;
Zero-coupon bond value:

$$P_{\text{bm}}(t,T) = a_{\text{bm}}(t,T)\exp[-b_{\text{bm}}(t,T)x_t]$$

where $b_{\text{bm}}(t,T) = b_{\text{cir}}(t,T)$ and

$$a_{\text{bm}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\frac{a_{\text{cir}}(0,t)e^{-b_{\text{cir}}(0,t)x_0}}{a_{\text{cir}}(0,T)e^{-b_{\text{cir}}(0,T)x_0}}a_{\text{cir}}(t,T)e^{\varphi(t)(T-t)}$$

All other formulas (options) follow CIR structure.

3.5 Black-Karasinski Model (arbitrage model)

Log-normal model:

$$d\ln(r_t) = \kappa(\varphi(t) - \ln(r_t))dt + \sigma dW_t$$

Key property: r_t is **log-normally distributed** and **always positive** (no negative rates). Equivalently, let $x_t = \ln(r_t)$; then x_t follows Vasicek dynamics with mean-reverting level $\varphi(t)$. Thus $r_t = \exp(x_t)$.
By Itô's lemma, dynamics of r_t:

$$\frac{dr_t}{r_t} = \left(\kappa\varphi(t) - \kappa\ln(r_t) + \frac{\sigma^2}{2}\right)dt + \sigma dW_t$$

Solution (exponential of Vasicek solution):

$$r_t = \exp\left[\ln(r_s)e^{-\kappa(t-s)} + \kappa\int_s^t \varphi(u)e^{-\kappa(t-u)}du + \sigma\int_s^t e^{-\kappa(t-u)}dW_u\right]$$

4 Multifactor Models

4.1 Two-Factor Gaussian Model

Dynamics of factors: Hull-White (1994) two-factor model (formulation by Brigo-Mercurio 2006). Two factors $\{x_t\}_{t\geq 0}$ and $\{y_t\}_{t\geq 0}$ with $x_0 = y_0 = 0$, model calibrated at $t = 0$ on yield curve:

$$dx_t = -ax_t dt + \sigma dW_{1t}$$

$$dy_t = -by_t dt + \eta dW_{2t}$$

where $\{W_{1t}\}$ and $\{W_{2t}\}$ are Brownian motions with correlation ρ. Instantaneous short rate:

$$r_t = x_t + y_t + \varphi(t)$$

Zero-coupon bond value:

$$P_{2fg}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp(B(t,T))$$

$$B(t,T) = \frac{1}{2}[V(t,T) - V(0,T) + V(0,t)] - \frac{1-e^{-a(T-t)}}{a}x_t - \frac{1-e^{-b(T-t)}}{b}y_t$$

European call: Same structure as one-factor model:

$$\text{call}_{2fg}(P,K) = P_{2fg}(0,T)\Phi(h) - KP_{2fg}(0,T_{\text{call}})\Phi(h-\sigma_p)$$

where $h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{2fg}(0,T)}{KP_{2fg}(0,T_{\text{call}})}$ and σ_p is volatility of bond returns (derived from $V(t,T)$ variance term).
Two-factor structure captures both level and slope of yield curve; better fit than one-factor models.

4.2 Fong-Vasicek Model

Two-factor extension of Vasicek: Uses rate volatility as second factor for interpretability. Dynamics under EMM:

$$dr_t = \kappa(\theta - r_t)dt + \sqrt{v_t}dZ_{1t}$$
$$dv_t = \gamma(\bar{v} - v_t)dt + \xi\sqrt{v_t}dZ_{2t}$$

where $\{Z_{1t}\}$ and $\{Z_{2t}\}$ are correlated Brownian motions with correlation ρ. First factor r_t = short rate; second factor v_t = variance of short rate.
Zero-coupon bond value: Exponential-affine structure:

$$P_{fvk}(t,T) = \exp[-b_{fvk,1}(t,T)r_t + b_{fvk,2}(t,T)v_t + b_{fvk,3}(t,T)]$$

with $b_{fvk,1}(t,T) = b_{vk}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$, while $b_{fvk,2}(t,T)$ and $b_{fvk,3}(t,T)$ involve confluent hypergeometric functions (require numerical integration; Selby-Strickland (1995) recursive algorithm available).
Second factor captures stochastic volatility of rates; improves fit relative to constant-volatility one-factor Vasicek.

4.3 Longstaff-Schwartz Model

Extension of CIR with stochastic volatility: Dynamics under EMM for short rate r_t and volatility v_t:

$$dr_t = \mu_r(r_t, v_t)dt + a\sqrt{\frac{\beta r_t - v_t}{a(\beta - a)}}dZ_{1t} + \beta\sqrt{\frac{v_t - ar_t}{\beta(\beta - a)}}dZ_{2t}$$

$$dv_t = \mu_v(r_t, v_t)dt + a^2\sqrt{\frac{\beta r_t - v_t}{a(\beta - a)}}dZ_{1t} + \beta^2\sqrt{\frac{v_t - ar_t}{\beta(\beta - a)}}dZ_{2t}$$

Key remarks: - Brownian motions $\{Z_{1t}\}$, $\{Z_{2t}\}$ are independent - Volatility constraint: $a\tau \leq v_t \leq b\tau$ - Parameter β must be sufficiently large relative to α for stochastic volatility behavior
Zero-coupon bond:

$$P_{ls}(t,T) = a_{ls,1}(t,T)^y a_{ls,2}(t,T)^z \exp[b_{ls,1}(t,T)r_t + b_{ls,2}(t,T)v_t + b_{ls,3}(t,T)]$$

European call:

$$\text{call}_{ls}(P,K) = P_{ls}(0,T)\chi^2(\cdot;\cdot) - KP_{ls}(0,T_{\text{call}})\chi^2(\cdot;\cdot)$$

4.4 Two-Factor Cox-Ingersoll-Ross Model (Brigo-Mercurio formulation)

$$dx_t = \kappa_1(\theta_1 - x_t)dt + \sigma_1\sqrt{x_t}dZ_{1t}$$
$$dy_t = \kappa_2(\theta_2 - y_t)dt + \sigma_2\sqrt{y_t}dZ_{2t}$$

where $\{Z_{1t}\}$ and $\{Z_{2t}\}$ are independent Brownian motions. Feller condition: $2\kappa_i\theta_i \geq \sigma_i^2$ for $i = 1, 2$ ensures $x_t, y_t \geq 0$.

Instantaneous risk-free rate:

$$r_t = x_t + y_t$$

Arbitrage-free version:

$$r_t = x_t + y_t + \varphi(t)$$

with shift $\varphi(t) = f^m(0,t) - f_1^{cir}(0,t) - f_2^{cir}(0,t)$. *Zero-coupon bond value:* Product of two one-factor CIR bond values:

$$P_{2fcir}(t,T) = a_{cir,1}(t,T)a_{cir,2}(t,T)\exp[-b_{cir,1}(t,T)x_t - b_{cir,2}(t,T)y_t]$$

with $a_{cir,i}(t,T) = a_{cir}(t,T)$ and $b_{cir,i}(t,T) = b_{cir}(t,T)$ from one-factor CIR model, replacing parameters $\{\kappa, \theta, \sigma\}$ with $\{\kappa_i, \theta_i, \sigma_i\}$ for $i = 1, 2$.

4.5 Multifactor Case

N-factor extension: Each factor follows CIR dynamics $dx_{it} = \kappa_i(\theta_i - x_{it})dt + \sigma_i\sqrt{x_{it}}dZ_{it}$ with independent Brownian motions; instantaneous rate $r_t = \sum_{i=1}^N x_{it}$ (or $+\varphi(t)$ for arbitrage-free version). Zero-coupon bond: product of N one-factor CIR bond values.

4.6 Affine Term Structure Models - Duffie-Kan

N exogenous factors $X_t = (X_{1t}, \ldots, X_{Nt})' \in \mathbb{R}^N$ follow under EMM:

$$dX_t = (K\theta - KX_t)dt + S_t dW_t$$

where K is $N \times N$ matrix, θ is $N \times 1$ vector, W_t is N-dimensional Brownian motion, S_t is $N \times N$ diagonal matrix with elements $[S_t]_{ii} = \sqrt{a_i + \beta_i' X_t}$ $[a_i \geq 0,\ \beta_i \in \mathbb{R}^N]$.
Market price of risk: Structure $\lambda_t = \lambda_1 S_t + S_t \lambda_2 X_t$ with $\lambda_1 \in \mathbb{R}^N$, λ_2 matrix of dimension $N \times N$.
Linear short rate:

$$r_t = \delta_0 + \delta_1' X_t$$

with $\delta_0 \in \mathbb{R}$, $\delta_1 \in \mathbb{R}^N$.
Zero-coupon bond (exponential-affine):

$$P_{aff}(t,T) = \exp[A(\tau) - B(\tau)'X_t]$$

with $\tau = T - t$ [time-to-maturity].

5 Implementation

Estimation vs Calibration: Estimation fits model to historical yield curve (ex-post analysis); calibration fits to contemporaneous market data (yields + derivatives prices) for precise point-in-time pricing. procidure: for equilibrium models, fit to yiled curve (minimizing squared errors); for arbitrage models, fit to yield curve then fit to caps/floors or swaption prices. we check the price fit but a more stringent test is to check the implied volatilities (using Blacks) fit.

5.1 Black Model for Caps and Floors

Black model approximation: Assumes forward rates are lognormal; ignores correlation between discount factor and payoff; recognizes future spot rate $\approx$ future forward rate $l_{t_{j-1}}(t_j) = F(t_{j-1}, t_j)$.
Cap price with strike K and maturity t_n:

$$\text{cap}(t_n) = M \sum_{j=i+1}^{n} \tau P(0, t_j) \left[F(t_{j-1}, t_j)\Phi(d_1) - K\Phi(d_2) \right]$$

where τ is accrual period (typically 0.25 for quarterly), M is notional, $P(0, t_j)$ is discount factor, and:

$$d_1 = \frac{1}{v_j} \left[\ln \frac{F(t_{j-1}, t_j)}{K} + \frac{v_j^2}{2} \right], \quad d_2 = d_1 - v_j, \quad v_j = \sigma_{\text{cap}} \sqrt{t_{j-1}}$$

with σ_{cap} = implied volatility from market quotes (typically varies by tenor; first reset at $t_0 = 0.25$, $t_1 = 0.5$, etc.).
Floor price: By put-call parity:

$$\text{floor}(t_n) = M \sum_{j=i+1}^{n} \tau P(0, t_j) \left[K\Phi(-d_2) - F(t_{j-1}, t_j)\Phi(-d_1) \right]$$

At-the-money (ATM) strike: Cap and floor are ATM when $K = K_f$ (forward swap rate):

$$K_f = \frac{P(0, t_a) - P(0, t_n)}{\sum_{j=i+1}^{n} \tau P(0, t_j)}$$

5.2 Black Model for Swaptions

Black approximation: Ignores correlation between discount factor and swaption payoff; assumes forward swap rates are driftless geometric Brownian motions under EMM.
Payer swaption price with strike K and maturity t_n:

$$\text{payer swaption}(t_n) = M \left(K_f \Phi(d_1) - K\Phi(d_2) \right) \sum_{j=a+1}^{n} \tau P(0, t_j)$$

Receiver swaption price:

$$\text{receiver swaption}(t_n) = M \left(K\Phi(-d_2) - K_f \Phi(-d_1) \right) \sum_{j=a+1}^{n} \tau P(0, t_j)$$

where K_f is forward swap rate, and:

$$d_1 = \frac{1}{v} \left[\ln \frac{K_f}{K} + \frac{v^2}{2} \right], \quad d_2 = \frac{1}{v} \left[\ln \frac{K_f}{K} - \frac{v^2}{2} \right]$$

$$v = \sigma_{\text{swapt}} \sqrt{t_a}$$

with σ_{swapt} = implied swaption volatility from market, t_a = start of swap period.

5.3 Kalman Filtering

State-space framework: Measurement equation relates observed zero-coupon yields to hidden state variables; transition equation describes state variable evolution; estimates parameters maximizing total log-likelihood.
Measurement equation:

$$ZCY_t = A + BX_t + \eta_t$$

where A is $N \times 1$ vector, B is $N \times K$ matrix, η_t is $N \times 1$ vector of measurement errors with diagonal covariance matrix $H = \text{diag}(h_{t_1}^2, \ldots, h_{t_N}^2)$ [variances may differ by maturity but constant over time; uncorrelated cross-sectionally].
Transition equation:

$$x_{i,t} = \mathbb{E}[x_{i,t}|x_{i,t-\delta}] + \sqrt{\text{Var}[x_{i,t}|x_{i,t-\delta}]}\xi_t$$

Algorithm: Loop over all observation dates: (1) predict next-period state estimators and error variance; (2) compute prediction error (observed yields − model-predicted yields) and its variance; (3) compute log-likelihood from prediction error; (4) update state estimators using prediction error and updated variance. Final parameters maximize total log-likelihood over sample.

5.4 Diebold-Li Approach

Dynamic extension of Nelson-Siegel smoothing.
with time-varying loadings $L_a = 1$, $L_b = \frac{1-e^{-\tau/\tau}}{T}$, $L_c = \left(\frac{1-e^{-\tau/\tau}}{T} - e^{-\tau/T}\right)$ [parameter τ fixed]. Factor representation:

$$zcy_t(T) = a_t L_a + b_t L_b + c_t L_c$$

where a_t = level, b_t = slope, c_t = curvature.
Forecasting: Set parameter τ such that curvature loading L_c reaches maximum at 30 months. Specify AR(1) dynamics for each factor:

$$\hat{x}_{t+h} = c_x + k_x \hat{x}_t, \quad \text{for } x = \{a, b, c\}$$

Parameter estimation: OLS regression of x_{t+h} on x_t estimates intercept c_x and slope k_x by minimizing squared residuals between observed future factor values and predicted values.

6 HJM and Market Models

HJM and Market Models

HJM General Framework

Heath, Jarrow & Morton (1992) model the entire yield curve by working directly with **instantaneous forward rates** $f(t, T)$ rather than the short rate r_t. The key insight: specifying the **forward rate volatility function** $\sigma(t, T)$ fully determines the dynamics of the whole yield curve. This opens a general family of term structure models.

Core result (HJM drift condition): Under the EMM, the drift of the instantaneous forward rate is *entirely pinned down* by its volatility — no free parameters remain for the drift. Formally, the forward rate dynamics are:

$$df(t,T) = \sigma(t,T) \int_t^T \sigma(t,u)\, du\, dt + \sigma(t,T)\, dW_t$$

Specifying $\sigma(t,T)$ completely characterizes the model for all maturities simultaneously.

Markov condition (Carverhill 1994): In general, $r_t = f(t,t)$ is *not* Markovian, making tree implementation impossible. The short rate is Markovian if and only if $\sigma(t,T) = h(t) \cdot g(T)$ (separable form). In that case the HJM model reduces to a **generalized Hull-White model** and can be implemented on a recombining tree.

Volatility specifications:

- **Ritchken–Sankarasubramanian**: $\sigma(t,T) = \psi_t \cdot \exp(-\kappa(T-t))$ with ψ_t stochastic; leads to a two-state Markov representation (r_t, ϕ_t) enabling tree/PDE implementation.
- **Li–Ritchken–Sankarasubramanian**: special case with $\sigma(t,T) = v_t^\beta \exp(-\kappa(T-t))$, $0 \le \beta \le 1$; interpolates between Gaussian ($\beta = 0$) and log-normal ($\beta = 1$) dynamics.
- **Mercurio–Moraleda / Moraleda–Vorst**: polynomial-exponential forms that generate a **humped term structure of volatility** (volatility peaks at intermediate maturities) — a stylized fact of cap implied vol surfaces. Former lacks closed-form ZCB; latter has closed-form ZCB.

Market Models (BGM/LMM)

Motivation: Market quotes cap/floor/swaption prices as Black implied vols, but the Black formula rests on invalid assumptions (ignores discount factor–payoff correlation; assumes driftless GBM for forward rates under the EMM). Market models *rigorously justify* the Black formula via a **change of probability measure**.

Key mechanism: The simply-compounded forward rate $F(t;T_1,T_2)$ is a ratio of traded assets divided by $P(t,T_2)$, so it is a T_2**-forward martingale** under the T_2-forward measure $\mathbb{Q}^{T_2}$. Under this measure it can be modeled as a driftless GBM, exactly recovering the Black caplet formula without approximations.

Cap Market Model (BGM/Miltersen–Sandmann–Sondermann/Jamshidian 1997):

- Each forward rate $F(t;T_{k-1},T_k)$ follows driftless GBM under its own forward measure $\mathbb{Q}^{T_k}$.
- Caplets are priced exactly with the Black formula; cap = sum of Black caplets.
- Calibration: fit the volatility function $\gamma(t,T_k)$ (e.g., parametric humped form $\gamma = [\alpha+\beta(T_k-t)]e^{-\delta(T_k-t)}+\theta$) to market cap implied vols by minimizing squared pricing errors.
- **Swaptions cannot be priced analytically** in the cap market model $\Rightarrow$ use Monte Carlo simulation.
- Under a different forward measure $\mathbb{Q}^{T_n}$, forward rates acquire a **drift correction**; simulation uses this drift-adjusted dynamics via Euler scheme.

Swap Market Model (Jamshidian 1997):

- The forward swap rate $K_f(t)$ is a ratio of traded assets, hence a martingale under the **swap measure** (annuity as numéraire). Models $K_f(t)$ as driftless GBM $\Rightarrow$ Black swaption formula is exact.

- **Cap/floor market model and swap market model are theoretically incompatible**: one cannot simultaneously use Black formula for both caps and swaptions in a single consistent model. Practitioners must choose one and price the other numerically.

Backward-Looking Rates (Post-LIBOR, Lyashenko–Mercurio 2019)

With the end of LIBOR, caps/floors/swaptions reference **compounded overnight rates set in arrears** (SOFR, CORRA). These are **backward-looking**: the rate for period $[T_1, T_2]$ is not known until T_2.

Key idea: Extend the zero-coupon bond definition beyond maturity: for $t > T$, set $P(t,T) = B(t)/B(T)$ (bank account ratio), which remains a valid numéraire. The backward-looking forward rate $\tilde{F}(t; T_1, T_2)$ is still a T_2-martingale, giving a formula analogous to the forward-looking case.

Volatility adjustment: Before T_1, $\tilde{F}(t; T_1, T_2)$ behaves like the forward-looking rate (same volatility $\gamma(t, T_2)$). Between T_1 and T_2, compounding of overnight rates induces a **linear volatility decay**: $\bar{\gamma}(t; T_1, T_2) = \min\left(1, \frac{T_2 - t}{T_2 - T_1}\right) \gamma(t, T_2)$. Market model structure and measure-change drift corrections carry over unchanged.

7 Numerical Methods

7.1 Binomial Trees

Heston (1995) binomial tree: Discrete-time arbitrage-free tree for one-factor model where short rate follows random walk with mean reversion. Short rate dynamics:

$$r_{t+\Delta t} = \alpha + \rho r_t + \sigma \varepsilon$$

Connection to continuous models: Equivalent to Vasicek or CIR with $\kappa = 1 - \rho$ and $\theta = \alpha/\kappa$.
No-arbitrage valuation: Portfolio value equals discounted expected contingent claim under EMM:

$$g(t) = e^{-r_t \Delta t}[\pi g_u(t + \Delta t) + (1 - \pi) g_d(t + \Delta t)]$$

with risk-neutral probability:

$$\pi = \frac{e^{r_t \Delta t} - d}{u - d}, \quad u = \exp[(-\alpha - \rho r_t - \sigma)\Delta t], \quad d = \exp[(-\alpha - \rho r_t + \sigma)\Delta t]$$

Ensures martingale property: discounted value process is martingale under EMM.

7.2 Trinomial Trees

Hull-White trinomial tree: Captures richer interest rate dynamics with faster convergence to continuous time limit; more complex than binomial trees but better for mean-reverting processes. Designed for Hull-White model (can adapt for log-normal processes like Black-Karasinski). Underlying process:

$$dx_t = -\kappa x_t dt + \sigma dW_t$$

Process is normally distributed and mean-reverting to zero.

Tree structure: At each time step Δt, discretized process value $x_{i,j}$ (where i = time index, j = space index) branches to three nodes with transition probabilities π_u, π_m, π_d: - Up state: $x_{i+1,j+1}$ - Middle state: $x_{i+1,j}$ - Down state: $x_{i+1,j-1}$
Probability constraints and moment matching: Transition probabilities sum to one and match first two moments of underlying process:

$$\pi_u + \pi_m + \pi_d = 1$$

$$\pi_u(x_{i,j} + \Delta x_j) + \pi_m x_{i,j} + \pi_d(x_{i,j} - \Delta x_j) = x_{i,j} e^{-\kappa \Delta t}$$

Variance equation:

$$\pi_u(x_{i,j} + \Delta x_j)^2 + \pi_m x_{i,j}^2 + \pi_d(x_{i,j} - \Delta x_j)^2 = [x_{i,j} + \Delta x_j(\pi_u - \pi_d)]^2 + \sigma^2 \frac{1 - e^{-2\kappa \Delta t}}{2\kappa}$$

Branching geometry: Tree geometry adapts in different regions to improve mean reversion capture and reduce negative rates: - **Process (A)**: Normal branching when $x_{i,j}$ neither too high nor too low; states $\{x_{i,j} - \Delta x, x_{i,j}, x_{i,j} + \Delta x\}$ - **Process (B)**: When $x_{i,j}$ is too high (pulls down); states $\{x_{i,j} - 2\Delta x, x_{i,j} - \Delta x, x_{i,j}\}$ - **Process (C)**: When $x_{i,j}$ is too low (pulls up); states $\{x_{i,j}, x_{i,j} + \Delta x, x_{i,j} + 2\Delta x\}$
Critical levels $j_{\min}$ and $j_{\max}$ determine where branching switches from (A) to (C) and from (A) to (B). Numbering states from bottom: $x_{i,j} = j\Delta x$; offsets chosen to ensure mean reversion and positivity.

7.3 Second Stage of the Tree: Calibration via Arrow-Debreu Prices

Objective: Calibrate tree-built nodes to match market zero-coupon bond prices using Arrow-Debreu prices Q_{ij} (state prices: value of claim paying \$1 at node j, time i).
Zero-coupon bond pricing constraint with shift parameter:

$$P_{NSS}(0, (i+1)\Delta t) = \sum_k Q_{ik} \exp(-\alpha_i \Delta t) \pi_k$$

Solve for shift parameter α_i to match observed bond price $P_{NSS}(0, (i+1)\Delta t)$ from market data.
State price propagation:

$$Q_{(i+1)j} = \sum_k Q_{ik} \pi_{kj} \exp\left(-(\alpha_i + k\Delta x)\Delta t\right)$$

where π_{kj} is transition probability from state k to state j.
Black-Karasinski extension: For log-normal rates, apply $f^{-1}(\cdot) = \exp(\cdot)$ and solve α_i numerically as the equation becomes nonlinear.
Key points: Match each bond maturity sequentially; one α_i per time step (one-factor); ensure positivity of rates if using normal models; use Newton-Raphson for implicit solutions.

7.4 Finite Difference Methods

Flexibility vs. convergence trade-off: The finite difference method is more flexible than trees because it can easily accommodate a wide variety of interest rate models. However, the method is slower to converge (at least with its simplest scheme) notably because it requires the discretization of an entire state-space.

7.4.1 The One-dimensional Explicit Scheme

One-factor diffusion model under EMM:

$$dr_t = \mu(t, r_t)dt + \sigma(t, r_t)dW_t$$

PDE for security value: In the absence of arbitrage, any fixed-income security can be viewed as a contingent claim on the instantaneous risk-free rate; its value $g(t, r_t)$ satisfies:

$$\frac{\partial g(t, r_t)}{\partial t} + \frac{\partial g(t, r_t)}{\partial r_t}\mu(t, r_t) + \frac{1}{2}\frac{\partial^2 g(t, r_t)}{\partial r_t^2}\sigma(t, r_t)^2 - r_t g(t, r_t) = 0$$

Grid discretization: A finite difference method consists in discretizing this PDE in a state-space represented by a grid with space step Δr and time step Δt. State nodes indexed by i (space, from bottom to top) and t (time, from current to maturity T).
Discretized Greeks: Let Δ_i^t, Γ_i^t and θ_i^t denote the discretized versions of the contingent claim's delta, gamma and theta respectively, all being evaluated at space node i (numbered from bottom to top) and time node t (numbered from current time 0 till maturity T). The drift $\mu(t, r_t)$ and volatility $\sigma(t, r_t)$ being functions of t and r_t, their discretized versions at space node i and time node t can be written as μ_i^t and σ_i^t. Thus, the discretized PDE is:

$$\theta_i^t + \Delta_i^t \mu_i^t + \frac{1}{2}\Gamma_i^t(\sigma_i^t)^2 - i\Delta r g = 0$$

Central difference approximations:

$$\Delta_i^t = \frac{g_{i+1}^t - g_{i-1}^t}{2\Delta r}$$

$$\Gamma_i^t = \frac{g_{i+1}^t - 2g_i^t + g_{i-1}^t}{(\Delta r)^2}$$

$$\theta_i^t = \frac{g_i^t - g_i^{t-1}}{\Delta t}$$

Backward recursion formula: Plugging the above approximations into the discretized PDE and rearranging:

$$g_i^{t-1} = A_i^t g_{i-1}^{t-1} + B_i^t g_i^{t-1} + C_i^t g_{i+1}^{t-1}$$

with $v_1 = \frac{\Delta t}{(\Delta r)^2}$, $v_2 = \frac{\Delta t}{\Delta r}$, and coefficients:

$$A_i^t = \frac{1}{2}[(\sigma_i^t)^2 v_1 - \mu_i^t v_2]$$

$$B_i^t = 1 - (\sigma_i^t)^2 v_1 - i\Delta r \Delta t$$

$$C_i^t = \frac{1}{2}[\mu_i^t v_2 + (\sigma_i^t)^2 v_1]$$

Key properties and stability:

- Sum of coefficients: $A_i^t + B_i^t + C_i^t = 1 - i\Delta r \Delta t$ captures the one-period discount factor; equation (7.25) embeds the martingale property of discounted price process.
- Weights A_i^t, B_i^t, C_i^t are not probabilities and may lie outside $[0, 1]$, potentially causing instability. Stability requires small v_1 and v_2: discretize time sufficiently relative to space (horizontally stretched grid).

- Choice of $r_{\max}$: balance between avoiding truncation bias (need $r_{\max}$ sufficiently high) and computational efficiency (higher $r_{\max}$ requires finer discretization). Trade-off between accuracy and cost.

Boundary conditions:

- Zero-coupon bonds: $g = 1$ when $r = 0$; $g = 0$ when $r = r_{\max}$.
- Deep in-the-money regions: linear approximation $g^t_{i_{\max}} = 2g^t_{i_{\max}-1} - g^t_{i_{\max}-2}$.
- Deep out-of-the-money regions: derivatives are worthless.
- American options: compute maximum of exercise and continuation values recursively at each node; use exercise value as boundary condition for deep in-the-money regions.

7.4.2 Two-dimensional Explicit Scheme

Two-factor diffusion model under EMM:

$$dx_s = \mu(t, x_s, y_t)dt + \sigma(t, x_s, y_t)dW^x_s$$

$$dy_t = \mu(t, x_s, y_t)dt + \sigma(t, x_s, y_t)dW^y_s$$

with $dW^x_s dW^y_s = \rho dt$ (correlation ρ between factors). Contingent claim value $g(t, x_s, y_t)$ satisfies:

$$\frac{\partial g(t,x_s,y_t)}{\partial t} + \frac{\partial g(t,x_s,y_t)}{\partial x_s}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g(t,x_s,y_t)}{\partial x_s^2}\sigma(t,x_s,y_t)^2 + \frac{\partial g(t,x_s,y_t)}{\partial y_t}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g(t,x_s,y_t)}{\partial y_t^2}\sigma$$

3-dimensional grid discretization: Space steps Δx and Δy; time step Δt. Notation: $g^t_{i,j}$ = contingent claim value at node i for state variable x (left to right), node j for state variable y (bottom to top), at time t.

Discretization approach: Apply central differences for partial derivatives and backward recursion as in one-dimensional case. Cross-partial derivative $\frac{\partial^2 g}{\partial x_s \partial y_t}$ discretized using four-point stencil. Solution proceeds backward through time on 2-D grid at each time step.

7.5 Monte Carlo Simulation

Generating normal random variables: Box-Muller method: given u_1, u_2 from uniform $[0, 1]$:

$$\varepsilon_1 = \sqrt{-2\ln(u_1)}\cos(2\pi u_2), \quad \varepsilon_2 = \sqrt{-2\ln(u_1)}\sin(2\pi u_2)$$

Euler discretization scheme:

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t$$

Full truncation scheme: To preserve non-negativity, replace r_t with $\max(r_t, 0)$ in drift and diffusion coefficients.

Milstein scheme: Higher accuracy via:

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t + \frac{\sigma}{4}\Delta t(\varepsilon_t^2 - 1)$$

Path-dependent valuation: For each simulated path j, compute discount factor as:

$$D^{(j)} = \exp\left(-\sum_{n=0}^{N-1} r_n^{(j)} \Delta t\right)$$

Then price security as:

$$\text{Price} = \mathbb{E}^Q[D^{(j)} \cdot X(T)] = \frac{1}{M}\sum_{j=1}^{M} D^{(j)} \cdot X^{(j)}(T)$$

where $X(T)$ is terminal payoff and M is number of paths.
Correlated Brownian motions: Generate independent $(\varepsilon_1, \varepsilon_2)$, then:

$$\Delta W_1 = \sqrt{\Delta t}\varepsilon_1, \quad \Delta W_2 = \sqrt{\Delta t}(\rho\varepsilon_1 + \sqrt{1-\rho^2}\varepsilon_2)$$

7.5.1 Path-Dependent Interest Rate Payoffs

Barrier caps/floors: Activated ("in"-type) or deactivated ("out"-type) first time reference rate reaches critical barrier level from below ("up") or above ("down"). Four main types: up-in cap, down-in floor, down-out cap, up-out floor. Payoff:

$$\left(i_{u-1}^{(u)} - K\right)^+ \cdot \mathbb{1}\left(\max_{s\in[u-1,u]} i_s^{(u)} > H\right)$$

Barrier condition applies to whole cap/floor: once a caplet activates, all subsequent caplets activate.
Bounded caps/floors: Total accumulated payout cannot exceed amount M. At each settlement, accumulated value is computed and test applied to determine if below limit:

$$\left(i_{u-1}^{(u)} - K\right)^+ \cdot \mathbb{1}\left(\sum_{j=1}^{u-1}\left(i_{j-1}^{(j)} - K\right)^+ < M\right)$$

Option deactivated first time test fails.
Incremental fixed swaps: Fixed leg calculated as weighted average of fixed and variable rates. For payer swap with notional M, fixed rate K, frequency τ:

$$-M\tau\left(\kappa + (1-x)u_t^{(u)}\right) + M\tau d_t^{(u)}$$

Fraction x contractually determined by reference rate tier: $x = 100\%$ if $i_t < i_1$; $x = 60\%$ if $i_2 \leq i_t \leq i_1$; $x = 0\%$ if $i_t > i_2$. At each reset, simulated rate determines fraction x and payoff.
Ratchet caps/floors: Strike adjusts to level of reference rate on preceding reset date. Payoff at settlement:

$$\left(i_{u-1}^{(u)} - i_{u-2}^{(u-1)}\right)^+$$

Simulation adjusted to compute difference between current and previous reset rates at each settlement date.
Rolling caps/floors: Nominal is not constant. When caplet (floorlet) not exercised (not in-the-money), its notional is added to next caplet (floorlet). At each reset date, apply test and adjust payoff accordingly.